



## Koneru Lakshmaiah Education Foundation

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### OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

#### Project Problem Statement Submission Form

##### Section / Team Details

Section No.: 2

Team No.: 13

Project Title: Linux Process Monitoring and Control System

##### Team Members

| Roll Number | Student Name         | Signature |
|-------------|----------------------|-----------|
| 2520030008  | Sagi Sahasra Srihasa |           |
| 2520030011  | Pamarthy Tejaswini   |           |
| 2520030609  | Vegi Siri Chandana   |           |

##### 1. Abstract

The **Linux Process Monitoring and Control System** is a Linux-based system programming project designed to monitor and manage processes running on a system. The project provides information about active processes, including process IDs (PIDs), parent process IDs (PPIDs), and resource-related details, while also allowing users to control processes through appropriate Linux mechanisms.

The system is implemented in C/C++ using Linux/POSIX system programming interfaces. It makes use of process management, Linux system calls, signals, file I/O, and the `/proc` filesystem to obtain process information and perform control operations. Users can monitor processes, search for specific processes, and perform operations such as sending signals to terminate or control a process.

The project focuses on demonstrating practical **Operating Systems and Systems Programming concepts** rather than developing a generic application. It provides a working environment for understanding how Linux represents processes, how processes can be monitored, and how operating-system mechanisms can be used to control their execution. The implementation will be tested using different process states and control operations, with results analyzed for correctness and system behavior.

The project aims to provide hands-on understanding of Linux process management and POSIX APIs through a functional command-line application.

## 2. Problem Statement

Linux systems run multiple processes simultaneously, making it necessary to monitor their status and manage their execution efficiently. This project aims to develop a Linux-based system that allows users to view information about running processes and perform basic process control operations. The system will use Linux/POSIX mechanisms such as the /proc filesystem, system calls, and signals to monitor and control processes. The project is relevant to Operating Systems and Systems Programming as it provides practical understanding of process management, process states, and communication between user programs and the operating system.

## 3. Objectives

1. To monitor and display information about active processes running in a Linux system.
2. To retrieve process details such as PID, PPID, process state, and other relevant information using the /proc filesystem.
3. To control processes using Linux system calls and signals, including terminating selected processes.
4. To demonstrate practical use of Linux/POSIX process management and system programming concepts.

## 4. Proposed Methodology

The system will be implemented as a **Linux-based command-line application using C/C++**. It will access the /proc filesystem to retrieve information about running processes such as PID, PPID, and process state. The user will be provided with options to view, search, and select processes for management. Linux system calls and signals will be used to perform control operations such as terminating selected processes. The system will continuously display updated process information when required, allowing users to monitor process activity and observe changes in process states. The implementation will be tested with different processes and control operations to verify correct system behavior.

## 5. Operating Systems Concepts / Linux APIs Used

*(List the relevant Operating Systems concepts, Linux system calls, POSIX APIs, or mechanisms that will be used.)*

| OS Concept / Linux API / System Call | Purpose in the Project                                                                                    |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------|
| /proc Filesystem                     | Retrieve information about running processes such as PID, PPID, process state, and other process details. |
| opendir() / readdir()                | Traverse /proc and identify directories corresponding to active processes.                                |
| kill() System Call                   | Send signals to selected processes for control operations such as termination.                            |
| Linux Signals (SIGTERM, SIGKILL)     | Control or terminate processes based on user selection.                                                   |
| Process Management                   | Monitor process states and perform basic management operations on running processes                       |

## 6. Individual Contribution

| Roll Number | Student Name         | Individual Responsibility |
|-------------|----------------------|---------------------------|
| 2520030008  | Sagi Sahasra Srihasa |                           |
| 2520030011  | Pamarthy Tejaswini   |                           |
| 2520030609  | Vegi Siri Chandana   |                           |

## 7. Tools / Platforms / Software Used

| Tool / Platform / Software | Purpose                                                                               |
|----------------------------|---------------------------------------------------------------------------------------|
| Linux / Ubuntu             | Development and execution environment for the Linux/POSIX system-programming project. |
| C / C++                    | Implement the process monitoring and control system.                                  |
| GDB                        | Debug the program and identify runtime or logic errors.                               |
| GCC / G++ Compiler         | Compile the C/C++ source code and generate the executable.                            |

## 8. Expected Outcome

The project is expected to provide a Linux-based system that can **monitor and display information about running processes** and allow users to perform basic **process control operations**. It will retrieve process details such as PID, PPID, and process state using the /proc filesystem and control selected processes using Linux signals and system calls. The system should successfully demonstrate process monitoring and management through a command-line environment, along with correct and reliable behavior during testing.

## 9. GitHub Repository

**GitHub Repository Link:** [https://github.com/sahasrasagi10/2520030008\\_011\\_609\\_OSSP](https://github.com/sahasrasagi10/2520030008_011_609_OSSP)

**Repository Created:** \_\_\_\_\_

☐ Yes      ☐ No

## Faculty Name and Signature

**Faculty Name:** \_\_\_\_\_

**Signature:** \_\_\_\_\_

**Remarks:** \_\_\_\_\_

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## Faculty Use

**Project Approved:** \_\_\_\_\_

☐ Yes      ☐ No

**Date:** \_\_\_\_\_

**Faculty Signature:** \_\_\_\_\_